Brain Memory

Persistent Memory for AI Coding Agents

npm: brain-memory
Status: Pre-release
github.com · npmjs.com

⚠ The Problem

AI coding agents start every session from scratch. Past decisions, project context, lessons learned, and user preferences are lost. Users re-explain the same things over and over. Existing solutions rely on opaque databases, cloud services, or flat key-value stores with no semantic structure.

✨ The Solution

Brain Memory gives your AI agent a **biologically-inspired memory system** that lives entirely on your file system. Memories are stored as readable Markdown files in a semantic directory hierarchy. The agent remembers, forgets, strengthens, and reorganizes knowledge — just like a real brain.

⚙ How It Works

1 Install in Seconds

`npx brain-memory`

Interactive setup for Claude Code, Gemini CLI, or Codex CLI. No config files needed.

2 Memories Form Naturally

Use `/brain:memorize` to capture decisions, insights, and context. The agent classifies and stores them in a semantic hierarchy automatically.

3 Recall & Evolve

Memories surface contextually via `/brain:remember`. Strength, associations, and structure evolve through use, sleep cycles, and spaced repetition.

🌟 Key Features

📁 File System = Database

Directory paths encode semantic meaning. No vector DB, no servers, no infrastructure. Pure file I/O.

🧠 Neuroscience-Grounded

11 peer-reviewed cognitive science concepts implemented as working algorithms, not metaphors.

🔒 Private & Local

Runs 100% locally. No cloud, no API calls. Human-readable YAML+Markdown, git-versioned.

🔄 Spreading Activation

Recalling one memory activates related ones via a weighted association graph — surfacing hidden connections.

🌙 Sleep Cycle

9-phase maintenance: replay, homeostasis, knowledge propagation, crystallization, reorganization, pruning, and creative dreaming.

⏳ Spaced Repetition

SM-2 algorithm schedules memory review. Each recall improves decay resistance — important memories become permanent.

🧠 Neuroscience Foundations

- **Ebbinghaus Decay** — Exponential forgetting with per-memory rates
- **Spreading Activation** — Collins & Loftus associative recall
- **Hebbian Learning** — "Neurons that fire together wire together"
- **Synaptic Homeostasis** — Tononi & Cirelli global scaling
- **Context-Dependent Recall** — Encoding specificity principle

🎯 Why Brain Memory?

✓ **Zero Dependencies**
No databases, servers, embeddings, or API calls. Just Node.js + your file system.

✓ **Human-Readable & Git-Friendly**
Every memory is a Markdown file. Browse with any editor. Track changes with git.

✓ **Bidirectional Lifecycle**
Not just storage — active maintenance, strengthening, reorganization, and pruning.

✓ **Multi-Agent Support**

- **Spacing Effect** — Distributed practice with diminishing returns
- **Episodic → Semantic** — Crystallization of events into knowledge
- **Memory Reconsolidation** — Active updating during sleep
- **Targeted Forgetting** — Forensic erasure with audit trail

Works with Claude Code, Gemini CLI, and OpenAI Codex CLI out of the box.

✓ **Surgical Erasure**
The `/brain:sunshine` command traces and removes all references across your entire brain.

10 Slash Commands

- `/brain:init` Create brain structure
- `/brain:memorize` Store new memories
- `/brain:remember` Contextual recall
- `/brain:review` Spaced repetition
- `/brain:explore` Visual tree browsing
- `/brain:consolidate` Merge weak memories
- `/brain:forget` Decay & archive
- `/brain:sunshine` Deep forensic erasure
- `/brain:sleep` 9-phase maintenance
- `/brain:status` Health dashboard

Get Started in 30 Seconds

One command. No configuration. Your AI agent never forgets again.

```
npx brain-memory
```

Node.js ≥ 18

Pre-release v0.1.0